

Hardware Documentation

Server — CyberPowerPC C Series ET8890-37125 (Tiamat @ 192.168.12.242 post-Proxmox)

Specs confirmed via SSH + PowerShell query on 2026-03-13

Component	Spec
CPU	AMD Ryzen 5 3600, 6-core / 12-thread, 3.6GHz base / 4.2GHz boost
RAM	32GB DDR4-3200 — two sticks, 2 slots free
GPU	XFX Radeon RX 580 Series, 4GB VRAM, Driver 31.0.21924.61
Motherboard	Gigabyte B450M DS3H WIFI-CF (MicroATX, AM4 socket)
Storage (OS)	240GB WD SSD
Storage (Media)	2TB WD HDD
Storage (External)	240GB Kingston SSD (USB) + 2GB USB flash drive
PSU	450W
Networking	Gigabit Ethernet + 802.11AC Wi-Fi (onboard)
OS	Proxmox 9.2.3
Build date	~March 2020
Chassis	Full tower, tempered glass side panel
Service number	888-937-5582

Recommended Upgrades

- RAM**  **UPGRADED:** 32GB DDR4-3200 (was 8GB, upgraded 2026-03-26)
 - VM-901 can now run alongside the full media stack
 - FlareSolvrr Cloudflare solving now possible (Chrome headless needs ~2GB)
 - B450M DS3H supports up to 128GB across 4 slots (2 slots used)
- OS:** Migrated to Proxmox VE 

3. **Storage:** 2TB WD HDD in use for media. 240GB WD SSD (sdb) passed through to VM-901 for games.
4. **Network:** Use wired Ethernet — avoid Wi-Fi for streaming stability
5. **USB NIC** ⌚ PENDING: TP-Link UE300 (USB-A 3.0, RTL8153 chipset, ~\$12) — needed to extend OpenWrt VM LAN to physical devices (phones, Fire TVs, laptop). Plug into any USB-A port (rear or top panel). Will become `vmbr2` bridge → OpenWrt LAN NIC → Archer AP switch → all physical devices get DHCP/DNS from OpenWrt+AdGuard. Until it arrives, OpenWrt serves CTs/VMs only.

GPU Passthrough Status

RX 580 (XFX) is **fully configured for VFIO passthrough** — no setup needed:

- `amd_iommu=on iommu=pt` in GRUB cmdline ✓
- IOMMU Group 2: `09:00.0` (GPU) + `09:00.1` (HDMI audio) — clean group ✓
- Both functions bound to `vfio-pci` driver ✓
- VFIO PCI IDs: `1002:67df,1002:aaf0` in `/etc/modprobe.d/vfio-pci.conf` ✓
- `vfio_iommu_type1 allow_unsafe_interrupts=1` set ✓

With GPU passed through, Proxmox host is **headless** — manage via web UI or SSH only.

GPU Assignment — VM-900 (Windows Gaming VM)

As of 2026-05-29 the RX 580 is dedicated to **VM-900** (Windows gaming VM):

- `/dev/dri` device passthrough **removed** from CT-300 LXC config (`/etc/pve/lxc/300.conf`)
- Jellyfin (CT-300) switched to **software decoding/encoding** —
`HardwareAccelerationType=none`, `EnableHardwareEncoding=false`,
`HardwareDecodingCodecs` cleared (`/etc/jellyfin/encoding.xml`)
- GPU is now free for exclusive VFIO passthrough to VM-900

To attach the GPU to VM-900 via Proxmox web UI: Hardware → Add → PCI Device → select `09:00.0` (and `09:00.1` for HDMI audio) with PCIe enabled.

Storage on Tiamat

Device	Size	Use
sda (ST2000DM008 HDD)	1.8TB	Proxmox LVM: root (96GB), media-hdd (1.5TB), all CT/VM disks
sdb (WDC WDS240G2G0A SSD)	223GB	Passed through to VM-901 (Windows games storage); contains existing Windows install on sdb4

Bahamut (Raspberry Pi 4, DietPi)

Bahamut is the current native edge DNS / reverse-proxy / password-manager node. The name "Ziggy" is no longer attached to hardware — Ziggy now refers to CT-900 (Open WebUI + SearXNG) running as an LXC on Tiamat.

Component	Spec
CPU	Broadcom BCM2711, quad-core Cortex-A72 @ 1.5 GHz (64-bit)
RAM	2 GB class Pi memory (observed 1.8 GiB usable)
Storage	microSD + optional USB SSD for swap (recommended)
Networking	Gigabit Ethernet + 802.11AC Wi-Fi (use wired)
OS	DietPi on Debian 13 (trixie)
IP	192.168.12.244 (static)
Services	Native AdGuard Home (:53 , :8081), native Caddy + DuckDNS (:80 , :443), native Vaultwarden (:8080 behind Caddy), DietPi Dashboard, TigerVNC

Docker, wg-easy, and WireGuard are removed from Bahamut. Vaultwarden is fronted by native Caddy for automatic TLS. AdGuard is the LAN primary resolver.

Laptop — Primary Admin / Dev Machine

Component	Spec
OS	Pop!_OS 22.04 LTS (KDE Plasma 5.24.7)
Kernel	6.17.9-76061709-generic (64-bit)
CPU	Intel Core i9-13900HX, 24-core (8P+16E) / 32-thread, 13th Gen
RAM	62.5 GB
GPU (integrated)	Intel Arc / Intel Xe Graphics (Mesa)
GPU (discrete)	NVIDIA GeForce RTX 4080 (laptop)
Graphics platform	X11
Role	Primary admin workstation, repo dev, Proxmox SSH, Android APK builds

NVIDIA RTX 4080 available for CUDA workloads — useful for AI/transcoding experiments.
Android APK builds run here (Java 17 + Android SDK).

Client Devices

Device	Role
2x Fire TV + Fire TV Cube	Jellyfin + Plex, Homarr/Overseerr via Silk Browser, nzb360 sideload, TiamatsStack APK
Android (tablet/phone)	Jellyfin, Plex, nzb360, Bitwarden
Laptop (i9-13900HX)	Proxmox admin, Homarr, MediaStack Control, SSH, APK builds

Network Requirements

- All devices on the same LAN for local streaming
- Server should have a static LAN IP (set via router DHCP reservation or Proxmox static config)
- Gigabit router/switch recommended for smooth 1080p/4K local streaming